KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 29 · MATH

The Tiny House Workshop — Teacher Lesson Plan & Answer Key

Learning objectives: Students interpret multiplication as scaling by predicting whether a product is greater than, less than or equal to a factor (comparing the other factor with 1), explain multiplying by a fraction equal to 1 as making an equivalent fraction, find the area of rectangles with fractional side lengths by tiling a unit square and by multiplying, find areas with mixed-number side lengths using four partial products or improper fractions, and solve multistep building problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Liora (10), who helps her uncle, Lisandro, at his tiny-house workshop in Austin, Texas. Ms. Linnea draws the floor plans, Mr. Lowen, the carpenter, cuts every board, and Liora's friend Lochlan comes by with questions of his own. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs, extra spaces and a final period.
2. Slide 1 — Warm-Up, Part 1: Decimals Review (4 pts, 2 each, Week 28 skill). If a student misses both warm-up slides, recount the decimal places in one product before moving on.
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): the bigger-or-smaller prediction rule, a decimal quotient, or a two-step pace problem.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W29.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why Tiny-House Builders Use Fractions: nonfiction connection about Austin, the capital of Texas, on the Colorado River of Texas. The International Residential Code's tiny-house section describes a tiny house as a home of 400 square feet or less, not counting lofts, so builders measure in fractions of a foot. Defines a square foot. No questions.
6. Slide 5 — Picture It: Multiplying as Scaling. Ms. Linnea resizes a 12-foot wall: 3/4 × 12 = 9 feet (factor less than 1), 1 × 12 = 12 feet, and 5/4 × 12 = 15 feet (factor greater than 1). Compare the factor with 1 before you multiply.
7. Slide 6 — Your Turn, Part 1: Predicting the Product (8 pts, 4 each): type greater or less, and choose the product that is less than, equal to, or greater than the number.
8. Slide 7 — Your Turn, Part 2: Scaling Twice (4 pts): scale once, then scale the answer again (a model house, a plan drawn at 1/2 inch per foot then copied at 3/5, or a porch and a step).
9. Slide 8 — Teach: How Much Bigger or Smaller? Mr. Lowen's shortcut: 1 1/4 × 12 = 12 + 3 = 15; 3/4 × 12 = 12 − 3 = 9; 1 1/2 × 10 = 15. Multiplying by a fraction equal to 1 gives an equivalent fraction (2/3 × 4/4 = 8/12), and 2/5 × 7/8 is less than 7/8.

10. Slide 9 — Your Turn, Part 1: How Much Bigger or Smaller? (8 pts, 4 each): find a product by adding or subtracting a fraction of the number, and choose an order, an equation, or a factor.
11. Slide 10 — Your Turn, Part 2: Scaling in the Workshop (4 pts): a main floor plus a loft 3/8 as large, a board cut twice, or the gallons left in a 60-gallon water tank.
12. Slide 11 — Teach: Area by Tiling: Fractional Side Lengths. A 3/4-foot by 2/3-foot shelf covers 6 of the 12 tiles in a unit square (each tile is 1/12 square foot), so 2/3 × 3/4 = 6/12 = 1/2 square foot. The denominators multiply to count the tiles in the whole square, and the numerators multiply to count the tiles in the shelf.
13. Slide 12 — Your Turn, Part 1: Area by Tiling (8 pts, 4 each): an area typed as a fraction (equivalent fractions and decimals are accepted), and a tile count.
14. Slide 13 — Your Turn, Part 2: Area by Tiling (4 pts): find one area, then double, triple or subtract it (loft windows, kitchen shelves, or a rug on 1 square yard).
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (scaling, factor, mixed number, equivalent fraction, unit square, tile, area, square foot).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Mixed-Number Side Lengths. Uncle Lisandro's 2 1/2-foot by 1 1/3-foot bench top: four partial products 2 + 1/2 + 2/3 + 1/6 = 3 1/3 square feet, or 5/2 × 4/3 = 20/6 = 3 1/3. Watch out: 2 × 1 and 1/2 × 1/3 are only two of the four parts. Scaling check: the area must be greater than 2 1/2.
18. Slide 17 — Your Turn: Mixed-Number Sides (6 pts, 3 each): a typed mixed-number area, and a choice about partial products, Lochlan's two-part mistake, or a scaling prediction.
19. Slide 18 — Real World: The Floor Plan (4 pts): a different situation in each set (loft floor not covered by a rug, the cost of $3-per-square-foot counter tile, or a porch design 3/5 as long).
20. Slide 19 — Real World: Building Day (4 pts): an 8 1/2-foot by 24-foot main floor under the 400-square-foot limit, two matching windows, or a model 1/24 as long and then 3/2 as long.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a "which explanation is correct?" item and a "find the mistake" item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (quotation marks around dialogue, capital letters, commas with a city and state, and one grammar fix such as their/they're or to/too). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences (a series, a renaming phrase, or an opening word with a comma, and a verb that agrees with its real subject). Both serial-comma and no-serial-comma forms of a list are accepted.
24. Slide 23 — Match-Up Round (10 pts): 5 products of 8 matched to their values, side lengths matched to areas, or words (scaling, unit square, area, mixed number, a fraction equal to 1) matched to meanings, 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick fraction questions against the student's own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: Move-In Day (8 pts, 4 each): a different situation in each set (an 8-foot by 20 1/2-foot main floor under the 400-square-foot rule, a garden bed with flowers in 3/4 of it, or 2 solar panels 5 1/4 feet by 3 1/3 feet): find the first area or expression, then finish the two-step answer.
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W29.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Tiny House Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Austin Tiny House Master Builder!" with the congratulations message "Your fraction plans passed every check in Austin!" and a big confetti burst; below 85 shows "Good Work at the Workshop!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: DECIMALS REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Multiplying decimals (Week 28): multiply 0.8 × 0.9. Product:	**0.72**	8 tenths × 9 tenths = 72 hundredths, so 0.8 × 0.9 = 0.72.
r2_s0	Set 1: Dividing by a decimal (Week 28): which division has the same quotient as 2.7 ÷ 0.3? [27 ÷ 3 / 27 ÷ 0.3 / 2.7 ÷ 3 / 270 ÷ 3]	**27 ÷ 3**	Multiply BOTH numbers by 10: 27 ÷ 3 = 9. The others change only one number or use different powers of 10.
r3_s0	Set 1: Bigger or smaller (Week 28): without computing, is 2.4 × 0.5 greater than or less than 2.4? Type greater or less.	**less** (also accepted: less than, smaller)	0.5 is less than 1, so the product is only part of 2.4. (It is 1.2.)
r4_s0	Set 1: Dividing by a decimal (Week 28): divide 7.5 ÷ 0.5. Quotient:	**15**	Multiply both by 10: 75 ÷ 5 = 15.
r1_s1	Set 2: Multiplying decimals (Week 28): multiply 3.5 × 1.2. Product:	**4.2** (also accepted: 4.20)	35 × 12 = 420, with two decimal places in all: 4.20 = 4.2. Estimate: 4 × 1 = 4.
r2_s1	Set 2: Dividing by a decimal (Week 28): what is 0.64 ÷ 0.08? [8 / 0.8 / 80 / 0.08]	**8**	Multiply both by 100: 64 ÷ 8 = 8. Check: 8 × 0.08 = 0.64.
r3_s1	Set 2: Two steps (Week 28): a runner finishes 3 kilometers in 12.6 minutes. At that pace, how many minutes would 2 kilometers take? Minutes:	**8.4** (also accepted: 8.4 minutes, 8.4 minute)	Pace: 12.6 ÷ 3 = 4.2 minutes per kilometer. Two kilometers: 4.2 × 2 = 8.4 minutes.
r4_s1	Set 2: Multiplying decimals (Week 28): multiply 0.6 × 0.4. Product:	**0.24**	6 tenths × 4 tenths = 24 hundredths = 0.24.
r1_s2	Set 3: Dividing by a decimal (Week 28): divide 5.4 ÷ 0.6. Quotient:	**9**	Multiply both by 10: 54 ÷ 6 = 9. Check: 9 × 0.6 = 5.4.
r2_s2	Set 3: Multiplying decimals (Week 28): what is 1.6 × 2.5? [4 / 40 / 0.4 / 3.1]	**4**	16 × 25 = 400, with two decimal places: 4.00 = 4. Estimate: 2 × 2.5 = 5, so 4 makes sense.
r3_s2	Set 3: Two steps (Week 28): find 3.6 ÷ 0.4, then multiply the quotient by 1.5. Answer:	**13.5**	3.6 ÷ 0.4 = 36 ÷ 4 = 9. Then 9 × 1.5 = 13.5.
r4_s2	Set 3: Bigger or smaller (Week 28): without computing, is 6 ÷ 0.3 greater than or less than 6? Type greater or less.	**greater** (also accepted: greater than, bigger)	0.3 is less than 1, so more than 6 groups of 0.3 fit. (The quotient is 20.)

YOUR TURN: PREDICTING THE PRODUCT (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Without multiplying: is 7/4 × 20 greater than or less than 20? Type greater or less.	**greater** (also accepted: greater than, greater than 20, bigger)	7/4 is greater than 1 (it is 1 3/4), so the product is greater than 20. (It is 35.)

ID	Question	Answer	Why (explain to student)
l2_s0	Set 1: Which product is LESS than 9? [2/3 × 9 / 4/3 × 9 / 3/2 × 9 / 9/9 × 9]	**2/3 × 9**	2/3 is less than 1, so 2/3 × 9 = 6 is less than 9. 4/3 and 3/2 are greater than 1, and 9/9 equals 1.
l1_s1	Set 2: Without multiplying: is 5/8 × 30 greater than or less than 30? Type greater or less.	**less** (also accepted: less than, less than 30, smaller)	5/8 is less than 1, so the product is only part of 30. (It is 18 3/4.)
l2_s1	Set 2: Which product is EQUAL to 14? [6/6 × 14 / 7/6 × 14 / 6/7 × 14 / 1/14 × 14]	**6/6 × 14**	6/6 = 1, and 1 × 14 = 14. 7/6 is greater than 1, 6/7 is less than 1, and 1/14 × 14 = 1.
l1_s2	Set 3: Without multiplying: is 1 1/3 × 15 greater than or less than 15? Type greater or less.	**greater** (also accepted: greater than, greater than 15, bigger)	1 1/3 is greater than 1, so the product is greater than 15. (It is 20.)
l2_s2	Set 3: Which product is GREATER than 2/3? [3/2 × 2/3 / 1/2 × 2/3 / 2/3 × 2/3 / 5/6 × 2/3]	**3/2 × 2/3**	3/2 is greater than 1, so 3/2 × 2/3 (which is 1) is greater than 2/3. The other factors are less than 1.

YOUR TURN: SCALING TWICE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l3_s0	Set 1: Two steps: Liora builds a model of a 24-foot tiny house at 1/12 of the real length. Then she builds a second model that is 3/2 as long as the first model. How many feet long is the second model? Feet:	**3** (also accepted: 3 feet, 3 foot)	First model: 1/12 × 24 = 2 feet. Second model: 3/2 × 2 = 3 feet.
l3_s1	Set 2: Two steps: on Ms. Linnea's plan, each foot of wall is drawn 1/2 inch long. A 20-foot wall is drawn on the plan. Then she makes a smaller copy of the plan at 3/5 of the size. How many inches long is the wall on the smaller copy? Inches:	**6** (also accepted: 6 inches, 6 inch)	Plan: 20 × 1/2 = 10 inches. Smaller copy: 3/5 × 10 = 6 inches.
l3_s2	Set 3: Two steps: the trailer under a tiny house is 8 feet wide. Uncle Lisandro builds a porch 1 1/2 times as wide as the trailer, and a step 1/3 as wide as the porch. How many feet wide is the step? Feet:	**4** (also accepted: 4 feet, 4 foot)	Porch: 1 1/2 × 8 = 8 + 4 = 12 feet. Step: 1/3 × 12 = 4 feet.

YOUR TURN: HOW MUCH BIGGER OR SMALLER? (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Find 1 1/4 × 16 by adding 16 + (1/4 × 16). Product:	**20**	1/4 of 16 is 4, so 1 1/4 × 16 = 16 + 4 = 20.
k2_s0	Set 1: Without multiplying, which list is in order from least to greatest? [2/3 × 6, then 6, then 4/3 × 6 / 6, then 2/3 × 6, then 4/3 × 6 / 4/3 × 6, then 6, then 2/3 × 6 / 2/3 × 6, then 4/3 × 6, then 6]	**2/3 × 6, then 6, then 4/3 × 6**	2/3 is less than 1, so 2/3 × 6 is less than 6. 4/3 is greater than 1, so 4/3 × 6 is greater than 6. (They are 4, 6, and 8.)
k1_s1	Set 2: Find 3/4 × 24 by subtracting 24 − (1/4 × 24). Product:	**18**	1/4 of 24 is 6, so 3/4 × 24 = 24 − 6 = 18.
k2_s1	Set 2: Which equation shows multiplying by a fraction equal to 1? [3/5 × 4/4 = 12/20 / 3/5 × 4/5 = 12/25 / 3/5 + 4/4 = 1 3/5 / 3/5 × 4 = 12/5]	**3/5 × 4/4 = 12/20**	4/4 = 1, so 3/5 × 4/4 = 12/20 is an equivalent fraction: 12/20 is the same amount as 3/5.

ID	Question	Answer	Why (explain to student)
k1_s2	Set 3: Find 1 1/2 × 14 by adding 14 + (1/2 × 14). Product:	**21**	1/2 of 14 is 7, so 1 1/2 × 14 = 14 + 7 = 21.
k2_s2	Set 3: Liora multiplies 7/8 by a number and gets a product GREATER than 7/8. Which number could she have used? [9/8 / 7/8 / 1/2 / 8/9]	**9/8**	Only a factor greater than 1 makes the product greater than 7/8. 9/8 is greater than 1, and the others are less than 1.

YOUR TURN: SCALING IN THE WORKSHOP (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k3_s0	Set 1: Two steps: the main floor of a tiny house is 240 square feet. The loft is 3/8 as large as the main floor. How many square feet are the main floor and the loft together? Square feet:	**330** (also accepted: 330 square feet, 330 square foot)	Loft: 3/8 × 240 = 90 square feet. Together: 240 + 90 = 330 square feet.
k3_s1	Set 2: Two steps: Mr. Lowen has a 10-foot board. He cuts off 2/5 of it for a shelf. Then he uses 3/4 of the piece that is left for a step. How long is the step? Feet:	**4 1/2** (also accepted, with or without the unit word: 4 2/4, 4 4/8, 4 8/16, 9/2, 4-1/2, 4 and 1/2, 4.5)	Shelf: 2/5 × 10 = 4 feet, so 6 feet are left. Step: 3/4 × 6 = 18/4 = 4 1/2 feet.
k3_s2	Set 3: Two steps: the water tank holds 60 gallons. A shower uses 1/4 of a full tank, and washing dishes uses 1/6 of a full tank. How many gallons are left? Gallons:	**35** (also accepted: 35 gallons, 35 gallon)	Shower: 1/4 × 60 = 15 gallons. Dishes: 1/6 × 60 = 10 gallons. Left: 60 − 15 − 10 = 35 gallons.

YOUR TURN: AREA BY TILING (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: A cutting board is 3/4 foot long and 1/3 foot wide. What is its area in square feet? Type a fraction. Square feet:	**1/4** (also accepted, with or without the unit word: 2/8, 4/16, 0.25, 3/12)	Area = length × width: 3/4 × 1/3 = 3/12 = 1/4 square foot. Tiles of 1/4 by 1/3 foot: 12 cover the square foot, and the board covers 3.
e2_s0	Set 1: A unit square is cut into tiles that are 1/5 foot by 1/3 foot. How many tiles cover the whole square foot? [15 / 8 / 5 / 3]	**15**	There are 5 columns and 3 rows of tiles, so 5 × 3 = 15 tiles. Each tile is 1/15 square foot.
e3_s0	Set 1: Two steps: each loft window is 3/4 yard wide and 2/3 yard tall. The loft has 2 windows. What is the total area of the windows in square yards? Square yards:	**1** (also accepted: 1 square yards, 1 square yard)	One window: 3/4 × 2/3 = 6/12 = 1/2 square yard. Two windows: 2 × 1/2 = 1 square yard.
e1_s1	Set 2: A window pane is 2/5 meter wide and 5/6 meter tall. What is its area in square meters? Type a fraction. Square meters:	**1/3** (also accepted: 1/3 square meters, 1/3 square meter, 10/30, 10/30 square meters)	Area: 2/5 × 5/6 = 10/30, which simplifies to 1/3 square meter.
e2_s1	Set 2: A rectangle that is 2/3 foot by 4/5 foot is tiled with tiles that are 1/3 foot by 1/5 foot. How many tiles cover the rectangle? [8 / 15 / 6 / 9]	**8**	2/3 foot holds 2 rows and 4/5 foot holds 4 columns, so 2 × 4 = 8 tiles. The whole square foot would hold 15, so the area is 8/15 square foot.
e3_s1	Set 2: Two steps: each kitchen shelf is 5/6 foot long and 3/5 foot deep. Mr. Lowen builds 3 shelves. What is the total area of the shelf tops in square feet? Square feet:	**1 1/2** (also accepted, with or without the unit word: 1 2/4, 1 4/8, 1 8/16, 3/2, 1-1/2, 1 and 1/2, 1.5)	One shelf: 5/6 × 3/5 = 15/30 = 1/2 square foot. Three shelves: 3 × 1/2 = 3/2 = 1 1/2 square feet.

ID	Question	Answer	Why (explain to student)
e1_s2	Set 3: A doormat is 5/6 yard long and 3/4 yard wide. What is its area in square yards? Type a fraction. Square yards:	**5/8** (also accepted: 5/8 square yards, 5/8 square yard, 10/16, 10/16 square yards, 10/16 square yard, 0.625, 0.625 square yards, 0.625 square yard, 15/24, 15/24 square yards)	Area: 5/6 × 3/4 = 15/24, which simplifies to 5/8 square yard.
e2_s2	Set 3: Tiles are 1/4 foot by 1/4 foot. How many tiles cover a square that is 3/4 foot on each side? [9 / 16 / 6 / 12]	**9**	3/4 foot holds 3 tiles each way, so 3 × 3 = 9 tiles. Each is 1/16 square foot, so the area is 9/16 square foot.
e3_s2	Set 3: Two steps: a rug that is 7/8 yard long and 2/3 yard wide lies on a floor space of exactly 1 square yard. How many square yards of the floor space are NOT covered? Square yards:	**5/12** (also accepted: 5/12 square yards, 5/12 square yard)	Rug: 7/8 × 2/3 = 14/24 = 7/12 square yard. Not covered: 1 − 7/12 = 5/12 square yard.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: “A factor greater than 1 makes the product greater than the number you started with.” Which factor makes 40 × ___ greater than 40? [6/5 / 5/6 / 1/2 / 4/4]	**6/5**	6/5 is greater than 1, so 6/5 × 40 = 48 is greater than 40. 5/6 and 1/2 are less than 1, and 4/4 equals 1.
v2_s0	Set 1: Type your answer: resizing a number by multiplying it by a factor is called ___. (one word)	**scaling**	Scaling resizes a number. A factor greater than 1 stretches it, and a factor less than 1 shrinks it.
v1_s1	Set 2: From the lesson: “A unit square is a square with sides 1 unit long, and its area is 1 square unit.” A unit square is cut into tiles that are 1/2 unit by 1/6 unit. What is the area of one tile? [1/12 square unit / 1/8 square unit / 2/6 square unit / 12 square units]	**1/12 square unit**	2 columns × 6 rows = 12 tiles fill the unit square, so each tile is 1/12 square unit. Also, 1/2 × 1/6 = 1/12.
v2_s1	Set 2: Type your answer: a whole number and a fraction together, like 2 1/3, is called a ___ number. (one word)	**mixed** (also accepted: mixed number)	A mixed number has a whole-number part and a fraction part.
v1_s2	Set 3: From the lesson: “Multiplying by a fraction equal to 1, such as 4/4, gives an equivalent fraction.” What is 5/6 × 3/3? [15/18 / 8/9 / 15/6 / 5/18]	**15/18**	3/3 = 1, so 5/6 × 3/3 = 15/18, an equivalent fraction that names the same amount as 5/6.
v2_s2	Set 3: Type your answer: length times width gives the ___ of a rectangle. (one word)	**area** (also accepted: the area)	The area of a rectangle is its length times its width, even with fractional sides.

YOUR TURN: MIXED-NUMBER SIDES (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: A cabinet door is 1 1/2 feet wide and 2 1/2 feet tall. What is its area in square feet? Square feet:	**3 3/4** (also accepted, with or without the unit word: 3 6/8, 3 12/16, 15/4, 3-3/4, 3 and 3/4, 3.75)	1 1/2 = 3/2 and 2 1/2 = 5/2. 3/2 × 5/2 = 15/4 = 3 3/4 square feet.

ID	Question	Answer	Why (explain to student)
t2_s0	Set 1: Which partial products are correct for 1 1/2 × 2 1/3? [2 + 1/3 + 1 + 1/6 / 2 + 1/6 / 2 + 1/3 + 1/2 + 1/6 / 3 + 1/6]	**2 + 1/3 + 1 + 1/6**	1 × 2 = 2, 1 × 1/3 = 1/3, 1/2 × 2 = 1, and 1/2 × 1/3 = 1/6. The total is 3 1/2.
t1_s1	Set 2: A loft hatch is 2 1/4 feet long and 1 1/3 feet wide. What is its area in square feet? Square feet:	**3** (also accepted: 3 square feet, 3 square foot)	2 1/4 = 9/4 and 1 1/3 = 4/3. 9/4 × 4/3 = 36/12 = 3 square feet.
t2_s1	Set 2: Lochlan says the area of a 2 1/2-foot by 3 1/2-foot table is 6 1/4 square feet, because 2 × 3 = 6 and 1/2 × 1/2 = 1/4. What is the correct area? [8 3/4 square feet / 6 1/4 square feet / 6 1/2 square feet / 7 1/4 square feet]	**8 3/4 square feet**	5/2 × 7/2 = 35/4 = 8 3/4. Lochlan left out 2 × 1/2 = 1 and 1/2 × 3 = 1 1/2.
t1_s2	Set 3: A wall panel is 3 1/3 feet wide and 1 1/2 feet tall. What is its area in square feet? Square feet:	**5** (also accepted: 5 square feet, 5 square foot)	3 1/3 = 10/3 and 1 1/2 = 3/2. 10/3 × 3/2 = 30/6 = 5 square feet.
t2_s2	Set 3: Without multiplying: is 2 3/4 × 1 1/5 greater than or less than 2 3/4? [Greater, because 1 1/5 is greater than 1 / Less, because 1/5 is a small fraction / Equal, because 1 1/5 is close to 1 / Less, because multiplying fractions always makes a smaller answer]	**Greater, because 1 1/5 is greater than 1**	The factor 1 1/5 is greater than 1, so the product is greater than 2 3/4. (It is 3 3/10.)

REAL WORLD: THE FLOOR PLAN (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation The sleeping loft is 6 1/2 feet long and 8 feet wide. A rug will cover 1/4 of the loft floor.* Two steps: how many square feet of the loft floor will NOT be covered by the rug? Square feet:	**39** (also accepted: 39 square feet, 39 square foot)	Loft: 6 1/2 × 8 = 48 + 4 = 52 square feet. Not covered: 3/4 × 52 = 39 square feet.
w1_s1	Set 2: *The Situation The kitchen counter is 4 1/2 feet long and 1 1/3 feet deep. The tile Ms. Linnea picked costs $3 for each square foot.* Two steps: how much will the tile for the counter cost? Dollars:	**18** (also accepted: 18 dollars, 18 dollar, 18.00, 18.00 dollars, 18.00 dollar)	Area: 9/2 × 4/3 = 36/6 = 6 square feet. Cost: 6 × $3 = $18.
w1_s2	Set 3: *The Situation The first porch design is 7 1/2 feet long and 4 feet wide. Ms. Linnea's second design keeps the same width but is only 3/5 as long.* Two steps: what is the area of the second porch design in square feet? Square feet:	**18** (also accepted: 18 square feet, 18 square foot)	New length: 3/5 × 7 1/2 = 3/5 × 15/2 = 45/10 = 4 1/2 feet. Area: 4 1/2 × 4 = 18 square feet.

REAL WORLD: BUILDING DAY (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Uncle Lisandro's trailer is 8 1/2 feet wide and 24 feet long. The main floor of the tiny house will cover the whole trailer. The tiny-house rule is 400 square feet or less, not counting lofts.* Two steps: how many square feet under the 400-square-foot limit is the main floor? [196 square feet / 204 square feet / 176 square feet / 296 square feet]	**196 square feet**	Floor: 8 1/2 × 24 = 192 + 12 = 204 square feet. Under the limit by 400 − 204 = 196 square feet.
w2_s1	Set 2: *The Situation Mr. Lowen is framing 2 matching windows. Each window is 2 1/4 feet wide and 2 2/3 feet tall.* Two steps: what is the total area of the two windows in square feet? Square feet:	**12** (also accepted: 12 square feet, 12 square foot)	One window: 9/4 × 8/3 = 72/12 = 6 square feet. Two windows: 2 × 6 = 12 square feet.
w2_s2	Set 3: *The Situation The real tiny house is 32 feet long. Liora builds a model that is 1/24 as long as the real house. Then Lochlan builds a bigger model that is 3/2 as long as Liora's model.* Two steps: how long is Lochlan's model? [2 feet / 1 1/3 feet / 48 feet / 3 feet]	**2 feet**	Liora: 1/24 × 32 = 32/24 = 1 1/3 feet. Lochlan: 3/2 × 4/3 = 12/6 = 2 feet.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct for why 5/6 × 12 is less than 12? [5/6 is less than 1, so the product is only part of 12 / Multiplying always makes a number smaller / 5/6 has a bigger denominator than numerator, so you subtract / 12 is a whole number, so any fraction makes it smaller]	**5/6 is less than 1, so the product is only part of 12**	A factor less than 1 shrinks the number: 5/6 × 12 = 10. A factor greater than 1, like 7/6, would make it bigger.
x2_s0	Set 1: Find the mistake: Lochlan says a tile that is 1/2 foot by 1/3 foot has an area of 2/5 square foot, because 1 + 1 = 2 and 2 + 3 = 5. What should he fix? [Multiply, do not add: 1/2 × 1/3 = 1/6 square foot / Nothing. 2/5 is correct / It should be 1/5 square foot / It should be 2/6 square foot]	**Multiply, do not add: 1/2 × 1/3 = 1/6 square foot**	Area is length times width. 6 tiles of this size fill a square foot, so each one is 1/6 square foot.
x1_s1	Set 2: Which explanation is correct for the area of a 3/5-foot by 2/3-foot shelf? [It covers 6 of the 15 tiles in a unit square, so the area is 6/15 = 2/5 square foot / It covers 5 tiles, because 3 + 2 = 5 / Its area is 5/8 square foot, because you add the tops and the bottoms / Its area is greater than 1 square foot, because it has two sides]	**It covers 6 of the 15 tiles in a unit square, so the area is 6/15 = 2/5 square foot**	5 columns by 3 rows makes 15 tiles in a square foot. The shelf covers 3 columns and 2 rows, which is 6 tiles: 6/15 = 2/5.

ID	Question	Answer	Why (explain to student)
x2_s1	Set 2: Find the mistake: a helper says 1 1/2 × 4 1/2 = 4 1/4, because 1 × 4 = 4 and 1/2 × 1/2 = 1/4. What should the helper fix? [Two partial products are missing; 3/2 × 9/2 = 27/4 = 6 3/4 / Nothing. 4 1/4 is correct / It should be 5 1/4 / It should be 6]	**Two partial products are missing; 3/2 × 9/2 = 27/4 = 6 3/4**	The missing parts are 1 × 1/2 = 1/2 and 1/2 × 4 = 2. 4 + 1/2 + 2 + 1/4 = 6 3/4.
x1_s2	Set 3: Liora multiplies 3/4 × 9/10 and says the product must be less than 3/4. Which explanation is correct? [She is right: 9/10 is less than 1, so she takes only part of 3/4 / She is wrong: 9/10 is close to 1, so the product equals 3/4 / She is wrong: multiplying makes numbers bigger / She is right: any product of two fractions is less than 1/2]	**She is right: 9/10 is less than 1, so she takes only part of 3/4**	Multiplying by a factor less than 1 makes the product smaller: 3/4 × 9/10 = 27/40, which is less than 30/40 = 3/4.
x2_s2	Set 3: Find the mistake: Lochlan says 8 × 3/2 = 12, so multiplying 8 by 3/2 made it smaller. What should he fix? [12 is greater than 8, because 3/2 is greater than 1 / Nothing. The product got smaller / The product should be 5 1/3 / The product should be 16/3]	**12 is greater than 8, because 3/2 is greater than 1**	3/2 = 1 1/2, which is greater than 1, so the product is bigger: 8 + 4 = 12.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: liora asked, how big is the loft	**Liora asked, "How big is the loft?"**	Capitalize the name Liora, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: We drove across austin, texas, to see the tiny houses.	**We drove across Austin, Texas, to see the tiny houses.**	Names of cities and states are proper nouns, so capitalize Austin and Texas.
s3_s0	Set 1: Fix this sentence: We bought wood nails and paint for the porch.	**We bought wood, nails, and paint for the porch.** (also accepted: We bought wood, nails and paint for the porch.)	Use commas to separate the items in a series: wood, nails, and paint.
s4_s0	Set 1: Fix this sentence: each of the windows have a screen	**Each of the windows has a screen.**	Capitalize the first word. The subject is each (one window at a time), so use has, not have. End with a period.
s1_s1	Set 2: Fix this sentence: ms. linnea said, measure twice	**Ms. Linnea said, "Measure twice."** (also accepted: Ms. Linnea said, "Measure twice!")	Capitalize Ms. and the name Linnea, put quotation marks around her exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: The builders put away they're tools.	**The builders put away their tools.**	Their shows ownership: the tools belong to the builders. They're means they are.
s3_s1	Set 2: Fix this sentence: Uncle Lisandro the head builder checked the plans.	**Uncle Lisandro, the head builder, checked the plans.**	The head builder describes Uncle Lisandro. Put a comma before and after a phrase like this that renames the person.

ID	Question	Answer	Why (explain to student)
s4_s1	Set 2: Fix this sentence: the boards in the stack is long	**The boards in the stack are long.**	Capitalize the first word. The subject is boards (plural), not stack, so use are, not is. End with a period.
s1_s2	Set 3: Fix this sentence: mr. lowen said, hand me the saw	**Mr. Lowen said, "Hand me the saw."** (also accepted: Mr. Lowen said, "Hand me the saw!")	Capitalize Mr. and the name Lowen, put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Lochlan wants to paint the door to.	**Lochlan wants to paint the door too.**	Too means also: Lochlan wants to paint the door, also. The first to (to paint) is correct.
s3_s2	Set 3: Fix this sentence: No I did not measure the door yet.	**No, I did not measure the door yet.**	Put a comma after an opening word like Yes, No, or Well.
s4_s2	Set 3: Fix this sentence: the walls of the house is green	**The walls of the house are green.**	Capitalize the first word. The subject is walls (plural), not house, so use are, not is. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 3/4 × 8	**6**	2 points if matched on the first try.
match_s0_2	Set 1: Match 5/4 × 8	**10**	2 points if matched on the first try.
match_s0_3	Set 1: Match 1/2 × 8	**4**	2 points if matched on the first try.
match_s0_4	Set 1: Match 8/8 × 8	**8**	2 points if matched on the first try.
match_s0_5	Set 1: Match 3/2 × 8	**12**	2 points if matched on the first try.
match_s1_1	Set 2: Match 1/2 foot by 1/2 foot	**1/4 square foot**	2 points if matched on the first try.
match_s1_2	Set 2: Match 1/4 foot by 2/3 foot	**1/6 square foot**	2 points if matched on the first try.
match_s1_3	Set 2: Match 1/3 foot by 3/5 foot	**1/5 square foot**	2 points if matched on the first try.
match_s1_4	Set 2: Match 1 1/2 feet by 2 feet	**3 square feet**	2 points if matched on the first try.
match_s1_5	Set 2: Match 2 1/2 feet by 2 feet	**5 square feet**	2 points if matched on the first try.
match_s2_1	Set 3: Match resizing by multiplying	**scaling**	2 points if matched on the first try.
match_s2_2	Set 3: Match a square with sides of 1 unit	**unit square**	2 points if matched on the first try.
match_s2_3	Set 3: Match length times width	**area**	2 points if matched on the first try.
match_s2_4	Set 3: Match 2 1/3	**mixed number**	2 points if matched on the first try.
match_s2_5	Set 3: Match 4/4 or 7/7	**a fraction equal to 1**	2 points if matched on the first try.

BOSS BATTLE: MOVE-IN DAY (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *The Final Floor Plan Ms. Linnea's final plan has a main floor that is 8 feet wide and 20 1/2 feet long. The tiny-house rule is 400 square feet or less, not counting lofts.* What is the area of the main floor? [164 square feet / 160 square feet / 28 1/2 square feet / 168 square feet]	**164 square feet**	8 × 20 1/2 = 8 × 20 + 8 × 1/2 = 160 + 4 = 164 square feet. (28 1/2 is the sum of the sides, not the area.)
b2_s0	Set 1: Two steps: how many more square feet could the main floor have and still be 400 square feet or less? Square feet:	**236** (also accepted: 236 square feet, 236 square foot)	Area: 8 × 20 1/2 = 164 square feet. Room left: 400 − 164 = 236 square feet.
b1_s1	Set 2: *The Garden Bed Liora builds a garden bed beside the tiny house. It is 3 1/2 feet long and 2 2/3 feet wide. She will plant flowers in 3/4 of the bed.* What is the area of the garden bed? [9 1/3 square feet / 6 1/3 square feet / 6 1/6 square feet / 9 2/3 square feet]	**9 1/3 square feet**	7/2 × 8/3 = 56/6 = 28/3 = 9 1/3 square feet. (6 1/6 is the sum of the sides, and 6 1/3 uses only 3 × 2 and 1/2 × 2/3.)
b2_s1	Set 2: Two steps: how many square feet of the bed will have flowers? Square feet:	**7** (also accepted: 7 square feet, 7 square foot)	Bed: 28/3 square feet. Flowers: 3/4 × 28/3 = 84/12 = 7 square feet.
b1_s2	Set 3: *The Solar Panels Mr. Lowen mounts 2 matching solar panels on the roof. Each panel is 5 1/4 feet long and 3 1/3 feet wide.* Which expression gives the area of ONE panel? [21/4 × 10/3 / 5 × 3 + 1/4 × 1/3 / 21/4 + 10/3 / 5/4 × 3/3]	**21/4 × 10/3**	Change each mixed number to an improper fraction: 5 1/4 = 21/4 and 3 1/3 = 10/3, then multiply. The first choice leaves out two partial products.
b2_s2	Set 3: Two steps: what is the total area of the 2 panels in square feet? Square feet:	**35** (also accepted: 35 square feet, 35 square foot)	One panel: 21/4 × 10/3 = 210/12 = 17 1/2 square feet. Two panels: 2 × 17 1/2 = 35 square feet.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W29.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: a) 0.6 × 0.9 b) 4.8 ÷ 0.6	a) 0.54 b) 48 ÷ 6 = 8
2	Is 3/4 × 20 more or less than 20?	Less, because 3/4 is less than 1, so the product is only part of 20 (it is 15).
3	2/3 × 3/5 in simplest form	6/15 = 2/5
4	Which number equals 1 1/2 × 8?	12 (8 + 4)
5	1 square foot split into 2 columns and 3 rows	Each small piece is 1/6 square foot (6 equal pieces). Check that the drawing shows 6 equal parts.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Write >, <, or = to compare with 18: a) 5/6 × 18 b) 9/7 × 18 c) 3/3 × 18	a) < (5/6 is less than 1) b) > (9/7 is greater than 1) c) = (3/3 equals 1)
2	Area: a) 3/4 foot by 2/5 foot b) 2 1/2 feet by 1 1/5 feet	a) 6/20 = 3/10 square foot b) 5/2 × 6/5 = 30/10 = 3 square feet (partial products: 2 + 2/5 + 1/2 + 1/10 = 3).
3	Main floor 8 by 22 1/2 feet; loft 2/5 as large; together	Main floor: 8 × 22 1/2 = 180 square feet. Loft: 2/5 × 180 = 72 square feet. Together: 252 square feet.
4	Lochlan says 2 1/2 × 2 1/2 = 4 1/4	Model: Lochlan found only two of the four partial products. He left out 2 × 1/2 = 1 and 1/2 × 2 = 1. The correct product is 4 + 1 + 1 + 1/4 = 6 1/4 (or 5/2 × 5/2 = 25/4 = 6 1/4).

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Tile a unit square to show 2/3 × 1/2; why the product is less than both factors	Cut the square into 3 rows and 2 columns (6 tiles); shade 2 rows and 1 column, so 2 of 6 tiles overlap: 2/6 = 1/3. It is less than 1/2 because we took only 2/3 of 1/2, and less than 2/3 because we took only 1/2 of 2/3; multiplying by a fraction less than 1 makes a smaller product.